File Recovery & Evidence Analysis using FTK Imager

Aim:

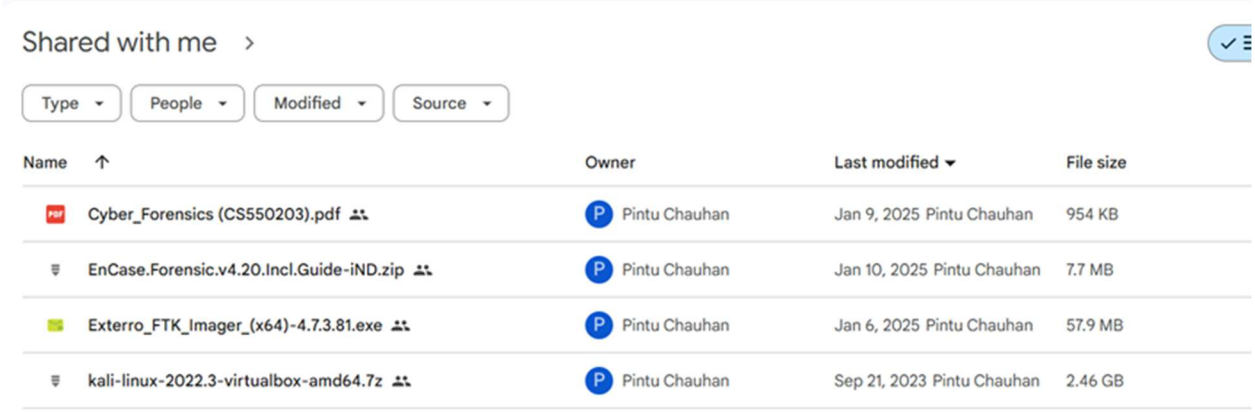
Install FTK_imager and conduct a file forensics investigation focusing on searching for files, analyzing log files, and recovering deleted files using Forensic Tool Kit (FTK). .

Requirements:

1. FTK_imager
2. Disk image for analysis (obtained legally for educational or training purposes).

Steps to Install FTK_Imager in Kali Linux:

1. **Download:**
   FTK_Imager in Kali Linux from link:
   https://drive.google.com/drive/folders/15hfVuzpTam3nSEHDOowXfQpw3GNAmQ3E



2. **Install wine:**
   Open a terminal window on Kali Linux. Run the following commands to install Wine:

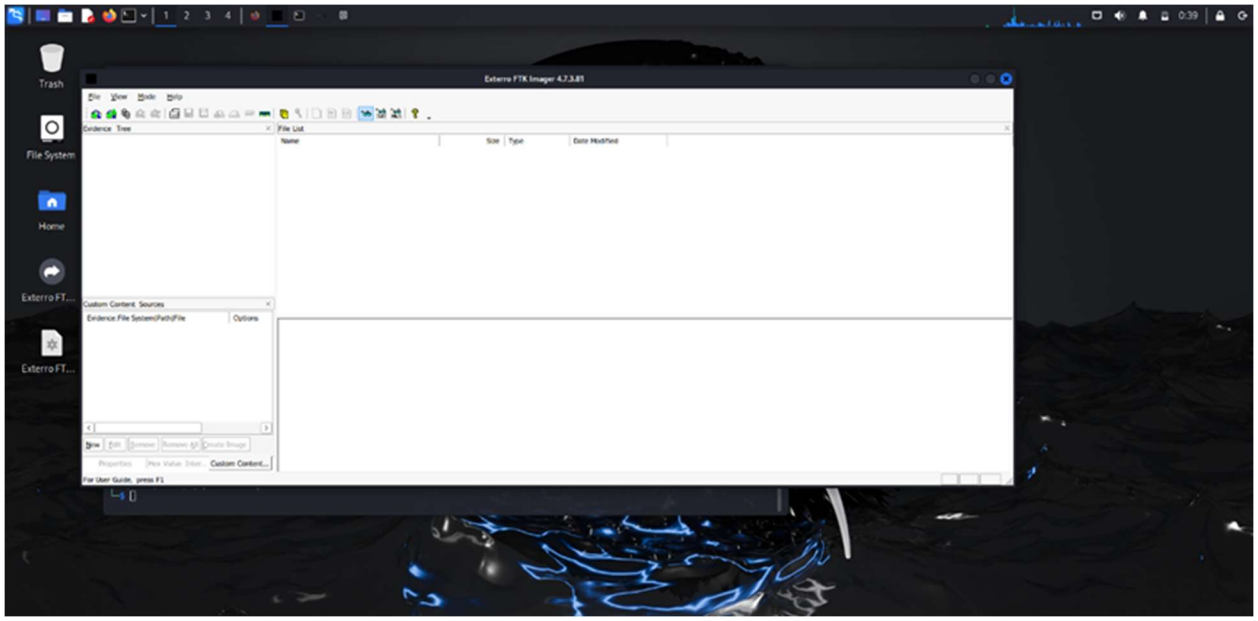
```
sudo dpkg --add-architecture i386
sudo apt update
sudo apt install wine64 wine32
```

3. **Install FTK:**
   Use Wine to run the installer. Replace setup.exe with the actual name of the FTK installer:

   wine setup.exe

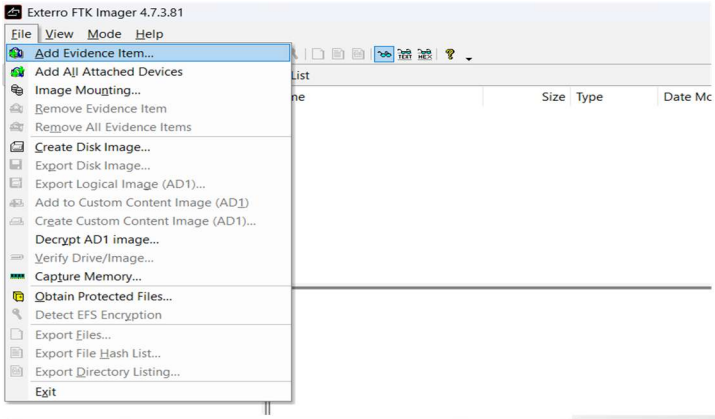
4. **Run FTK:**
   Once the installation is complete, attempt to run FTK using Wine:
   wine "/path/to/FTK/FTK.exe"

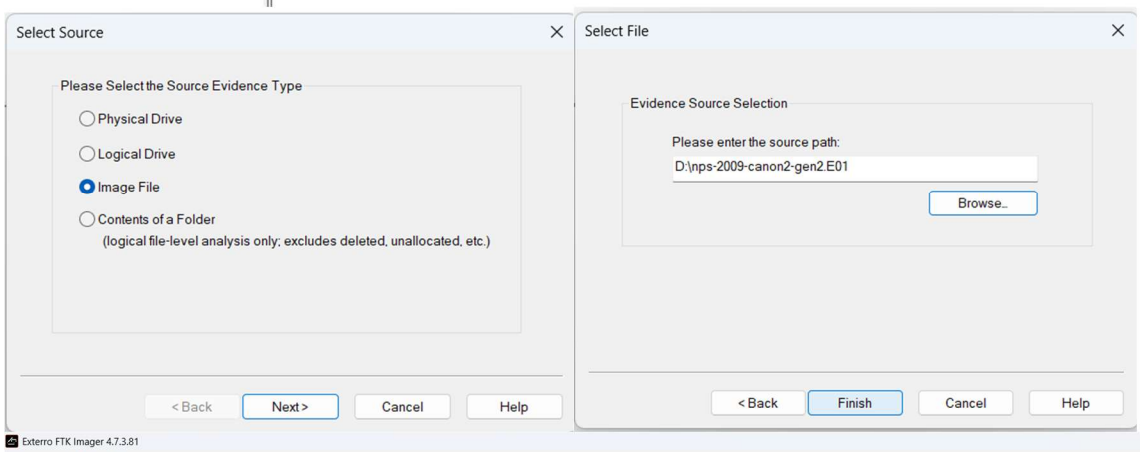
   Replace /path/to/FTK/ with the actual path to the FTK executable.

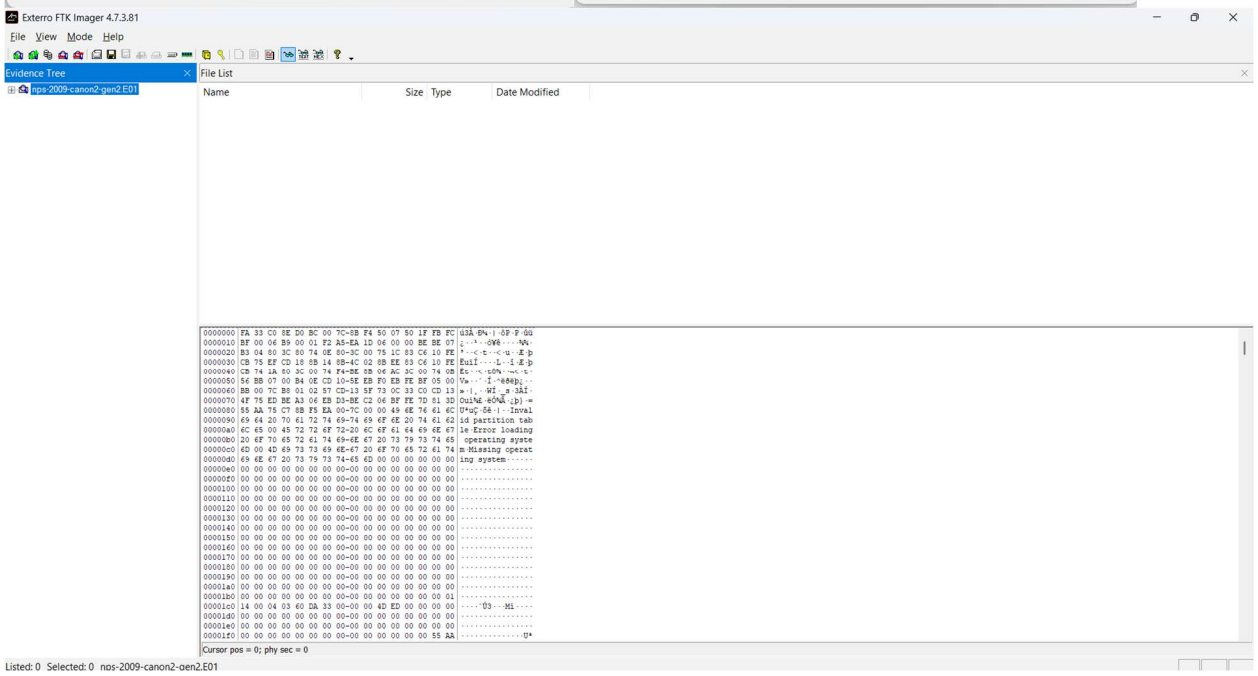


Steps to perform analysis of Disk Image:

1. **Import Disk Image into FTK:** Launch FTK and create a new case. Import the disk image into the case for analysis.

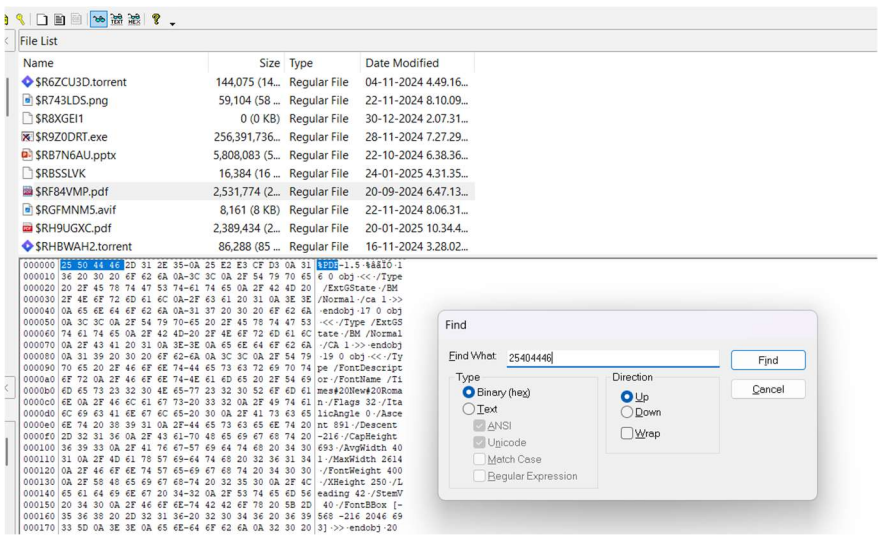


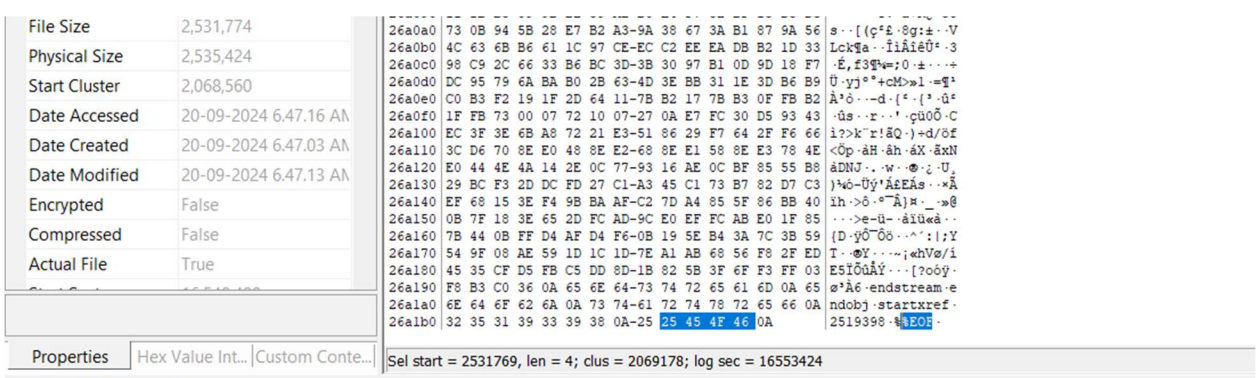




2. **Search for Files**:

- Load the Disk Image
- Open FTK and load the disk image.
- Allow FTK to process and index the files.
- Perform a Hexadecimal Search for the PDF Signature
- Navigate to the Search tab in FTK.
- Select Hexadecimal Search.
- Enter the hex pattern **25 50 44 46** to search for files beginning with %PDF.
- Execute the search and review the results.
- Validate Identified Files Using EOF Marker
- After locating files with the PDF signature, conduct another hex search using **25 45 4F 46** (%%EOF).
- Confirm that these markers appear at the end of the suspected files.
- If both markers are present, the file is most likely a valid PDF.



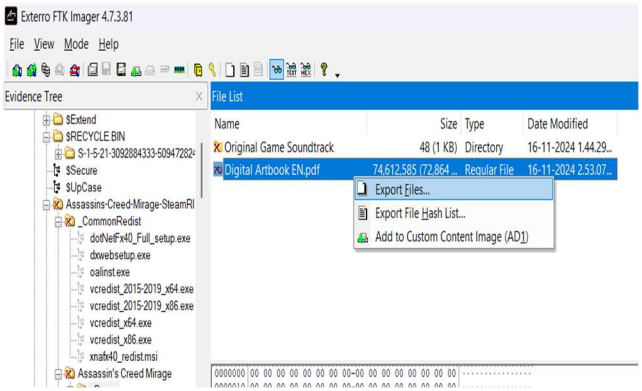


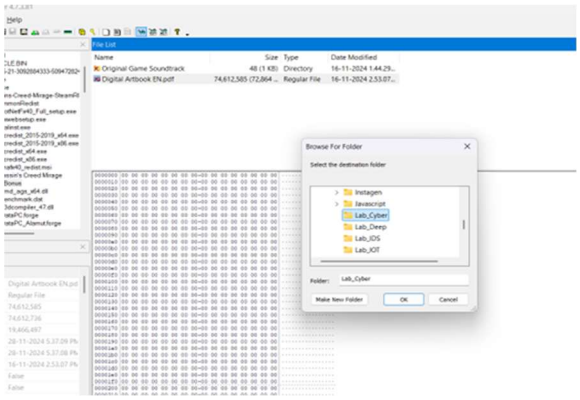
3. **Analyze Log Files:** Use FTK to locate and analyze system log files within the disk image. Investigate timestamps, events, and any suspicious activities within the log files. Document findings related to the log files.
    - Click File in FTK

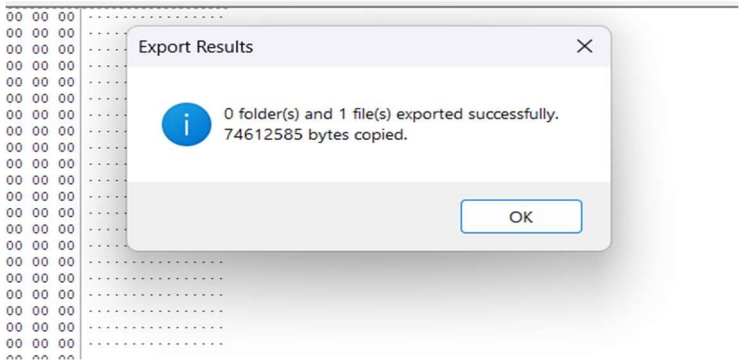
- Create a disk image
- Select your image file and browse the path
- Select the image destination type (Select Raw(dd))
- Fill Evidence number, case no., etc
- Provide image destination and file name you want to save
- Then select an option *Create directory listings of all files in the image after they are created*.

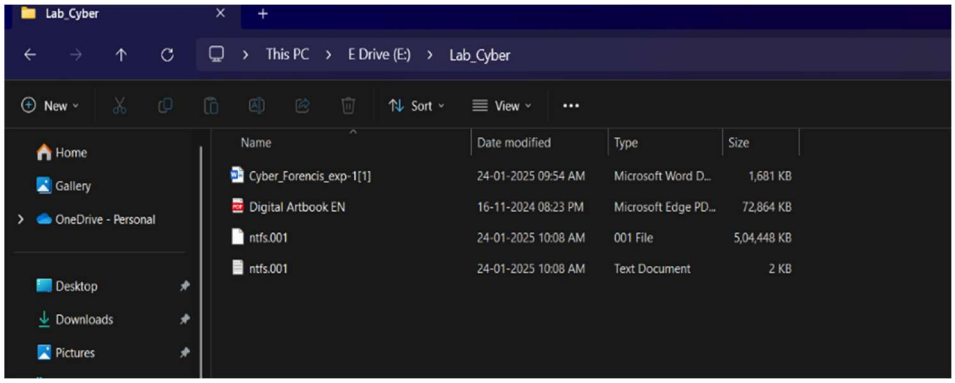
A	B	C	D	E	F	G	H	I	J
Filename	Full Path	Size (bytes)	Created	Modified	Accessed	Is Deleted			
[root]	Partition 1\CANON_DC [FAT12]\[root]\	16384				no			
VBR	Partition 1\CANON_DC [FAT12]\VBR	512				no			
[unallocated space]	Partition 1\CANON_DC [FAT12]\[unallocated s	0				no			
FAT1	Partition 1\CANON_DC [FAT12]\FAT1	3072				no			
FAT2	Partition 1\CANON_DC [FAT12]\FAT2	3072				no			
DCIM	Partition 1\CANON_DC [FAT12]\[root]\DCIM\	16384	2008-Dec-23 14:12:38	2008-Dec-23 14:12:38		no			
100CANON	Partition 1\CANON_DC [FAT12]\[root]\DCIM\1	16384	2008-Dec-23 14:12:38	2008-Dec-23 14:12:38		no			
IMG_0001.JPG	Partition 1\CANON_DC [FAT12]\[root]\DCIM\1	855935	2008-Dec-23 14:12:38	2008-Dec-23 14:12:38		no			
IMG_0002.JPG	Partition 1\CANON_DC [FAT12]\[root]\DCIM\1	871587	2008-Dec-23 14:12:46	2008-Dec-23 14:12:46		no			
IMG_0003.JPG	Partition 1\CANON_DC [FAT12]\[root]\DCIM\1	840101	2008-Dec-23 14:12:52	2008-Dec-23 14:12:52		no			
IMG_0004.JPG	Partition 1\CANON_DC [FAT12]\[root]\DCIM\1	859050	2008-Dec-23 14:12:56	2008-Dec-23 14:12:56		no			
IMG_0005.JPG	Partition 1\CANON_DC [FAT12]\[root]\DCIM\1	836531	2008-Dec-23 14:13:02	2008-Dec-23 14:13:02		no			
IMG_0006.JPG	Partition 1\CANON_DC [FAT12]\[root]\DCIM\1	777709	2008-Dec-23 14:13:08	2008-Dec-23 14:13:08		no			
IMG_0007.JPG	Partition 1\CANON_DC [FAT12]\[root]\DCIM\1	865313	2008-Dec-23 14:13:12	2008-Dec-23 14:13:12		no			
IMG_0008.JPG	Partition 1\CANON_DC [FAT12]\[root]\DCIM\1	745065	2008-Dec-23 14:13:16	2008-Dec-23 14:13:16		no			
IMG_0009.JPG	Partition 1\CANON_DC [FAT12]\[root]\DCIM\1	840692	2008-Dec-23 14:13:22	2008-Dec-23 14:13:22		no			
IMG_0010.JPG	Partition 1\CANON_DC [FAT12]\[root]\DCIM\1	853643	2008-Dec-23 14:13:26	2008-Dec-23 14:13:26		no			
IMG_0011.JPG	Partition 1\CANON_DC [FAT12]\[root]\DCIM\1	771052	2008-Dec-23 14:13:30	2008-Dec-23 14:13:30		no			
IMG_0012.JPG	Partition 1\CANON_DC [FAT12]\[root]\DCIM\1	815821	2008-Dec-23 14:13:36	2008-Dec-23 14:13:36		no			
IMG_0013.JPG	Partition 1\CANON_DC [FAT12]\[root]\DCIM\1	842160	2008-Dec-23 14:13:40	2008-Dec-23 14:13:40		no			
IMG_0014.JPG	Partition 1\CANON_DC [FAT12]\[root]\DCIM\1	873437	2008-Dec-23 14:13:44	2008-Dec-23 14:13:44		no			
IMG_0015.JPG	Partition 1\CANON_DC [FAT12]\[root]\DCIM\1	821758	2008-Dec-23 14:13:52	2008-Dec-23 14:13:52		no			
IMG_0016.JPG	Partition 1\CANON_DC [FAT12]\[root]\DCIM\1	853839	2008-Dec-23 14:13:56	2008-Dec-23 14:13:56		no			
IMG_0017.JPG	Partition 1\CANON_DC [FAT12]\[root]\DCIM\1	795574	2008-Dec-23 14:14:00	2008-Dec-23 14:14:00		no			
IMG_0018.JPG	Partition 1\CANON_DC [FAT12]\[root]\DCIM\1	784455	2008-Dec-23 14:14:06	2008-Dec-23 14:14:06		no			
IMG_0019.JPG	Partition 1\CANON_DC [FAT12]\[root]\DCIM\1	864257	2008-Dec-23 14:14:10	2008-Dec-23 14:14:10		no			
IMG_0020.JPG	Partition 1\CANON_DC [FAT12]\[root]\DCIM\1	883127	2008-Dec-23 14:14:16	2008-Dec-23 14:14:16		no			

4. **Recover Deleted Files:** Utilize FTK's file recovery features to identify and recover deleted files from the disk image. Explore options for filtering and sorting recovered files. Document the recovery process and the recovered files









5. **Generate Reports:** Generate comprehensive reports documenting the findings from the file search, log file analysis, and deleted file recovery. Customize the reports to include relevant metadata and analysis detail

File Name	File Path	File Signature Found	EOF Marker Found	Status
$RF84VMP.pdf	/root/$RECYCLEBIN/	Yes (`25 50 44 46`)	Yes (`25 45 4F 46`)	Valid PDF
Digital Artbook EN.pdf	/Assassin's creed Mirage/Bonus	No	No	Corrupt/Incomplete
unknown.bin	/Unallocated Space/	Yes (`25 50 44 46`)	No	Possible Fragment